

Learning4help by Waqas

Asslam-o-Alaikum! In this file we'll provide you **CS301P Important Final term MCQs. Must Prepare before Final term Exams.** I am not responsible for any answer. Verify Answers yourself too. If you found mistake then inform me.

If you found mistake then let us know.

You can contact us on WhatsApp <https://wa.me/+923421718143>

Or you can send us Gmail: Learning4help@gmail.com



**LEARNING
4 HELP**

Subscribe to my Channel:

https://www.youtube.com/channel/UC_CGyvvKgSGACQHPuq-951Q

VIRTUAL UNIVERSITY OF PAKISTAN

SERVICES:

- Assignments
- Quizzes
- GDBs
- LMS Handling
- Projects Making
- projects preparation
- online Tuition

We always do our best in serving our students and give them satisfied impressions.



**LEARN4
HELP**



CALL US NOW!



0342-1718143



www.learning4help.com

LT [Icons] vulms.vu.edu.pk/Quiz/QuizQuestion.aspx [Icons]

Handouts.pdf Login - Exams MTH603 Mid Term... CS601 Mcqs_Combi... plantuml.com/sitem... Phonetics and Phon... ENG-508 Handouts... Teaching of Listenin... Go to Extensions

The extension upgrade requires additional permissions. Go to the extensions manager to accept the upgrade.

CS301P - Data Structures (Practical) (Graded Quiz - 3) Quiz Start Time: 08:28 PM, 30 December 2024

Question # 1 of 10 (Start time: 08:28:44 PM, 30 December 2024) Total Marks: 1

If the data is given in sorted order, the tree generated will be similar to_____.

Select the correct option

☐ Stack

☒ Linked list

☐ Queue

☐ Heap

Reload Math Equations

Search [Icons] vulms.vu.edu.pk/Quiz/QuizQuestion.aspx [Icons]

Handouts.pdf Login - Exams MTH603 Mid Term... CS601 Mcqs_Combi... plantuml.com/sitem... Phonetics and Phon... ENG-508 Handouts... Teaching of Listenin... Go to Extensions

The extension upgrade requires additional permissions. Go to the extensions manager to accept the upgrade.

CS301P - Data Structures (Practical) (Graded Quiz - 3) Quiz Start Time: 08:28 PM, 30 December 2024

Question # 2 of 10 (Start time: 08:29:07 PM, 30 December 2024) Total Marks: 1

Which of the following function prototype is declaring the function as constant?

Select the correct option

☐ const EType& findMin();

☐ void EType& findMin(const int a);

☐ EType& findMin(const int a);

☒ const EType& findMin() const;

Reload Math Equations

LT [Icons] vulms.vu.edu.pk/Quiz/QuizQuestion.aspx [Icons]

Handouts.pdf Login - Exams MTH603 Mid Term... CS601 Mcqs_Combi... plantuml.com/sitem... Phonetics and Phon... ENG-508 Handouts... Teaching of Listenin... Go to Extensions

The extension upgrade requires additional permissions. Go to the extensions manager to accept the upgrade.

CS301P - Data Structures (Practical) (Graded Quiz - 3) Quiz Start Time: 08:28 PM, 30 December 2024

Question # 3 of 10 (Start time: 08:29:36 PM, 30 December 2024) Total Marks: 1

An AVL tree is identical to _____

Select the correct option

☒ Binary Search Tree

☐ Linked list

☐ Queue

☐ Stack

Reload Math Equations

Search [Icons] vulms.vu.edu.pk/Quiz/QuizQuestion.aspx [Icons]

Handouts.pdf Login - Exams MTH603 Mid Term... CS601 Mcqs_Combi... plantuml.com/sitem... Phonetics and Phon... ENG-508 Handouts... Teaching of Listenin... Go to Extensions

The extension upgrade requires additional permissions. Go to the extensions manager to accept the upgrade.

CS301P - Data Structures (Practical) (Graded Quiz - 3) Quiz Start Time: 08:28 PM, 30 December 2024

Question # 4 of 10 (Start time: 08:29:54 PM, 30 December 2024) Total Marks: 1

If values 9,5,7 are used to build AVL tree then which type of rotation can balance the AVL tree?

Select the correct option

☐ Single right

☐ Double left-right

☒ Double right-left

☐ Single left

Reload Math Equations

Learning4Help By Waqar Qureshi VULMS Handling 03421718143

BC230402652: PHOOL BIBI

CS301P - Data Structures (Practical) (Graded Quiz - 3)

Time Left: 25 sec(s)

Quiz Start Time: 08:28 PM, 30 December 2024

Question # 6 of 10 (Start time: 08:31:46 PM, 30 December 2024)

When type of rotation can balance the following AVL tree?

```
graph TD; 11((11)) --- 10((10)); 11 --- 16((16)); 10 --- 12((12));
```

Select the correct option

- ☐ Single left
- ☐ Single right
- ☐ Double right-left
- ☒ Double left-right

Question # 7 of 10 (Start time: 08:33:05 PM, 30 December 2024)

What will be the height of an empty AVL tree?

Select the correct option

- ☐ 2
- ☒ -1
- ☐ 0
- ☐ 1

LT [Icons] vulms.vu.edu.pk/Quiz/QuizQuestion.aspx [Icons]

Handouts.pdf Login - Exams MTH603 Mid Term... CS601 Mcqs_Combi... plantuml.com/sitem... Phonetics and Phon... ENG-508 Handouts... Teaching of Listenin... Go to Extensions

The extension upgrade requires additional permissions. Go to the extensions manager to accept the upgrade.

CS301P - Data Structures (Practical) (Graded Quiz - 3) Quiz Start Time: 08:28 PM, 30 December 2024

Question # 8 of 10 (Start time: 08:34:04 PM, 30 December 2024) Total Marks: 1

In the given BST, the balance factor of root node is:

Select the correct option

☐ 1

☐ -2

☒ 0

☐ -1

Reload Math Equations

Click to Save Answer & Move to Next Question

LT [Icons] vulms.vu.edu.pk/Quiz/QuizQuestion.aspx [Icons]

Handouts.pdf Login - Exams MTH603 Mid Term... CS601 Mcqs_Combi... plantuml.com/sitem... Phonetics and Phon... ENG-508 Handouts... Teaching of Listenin... Go to Extensions

The extension upgrade requires additional permissions. Go to the extensions manager to accept the upgrade.

CS301P - Data Structures (Practical) (Graded Quiz - 3) Quiz Start Time: 08:28 PM, 30 December 2024

Question # 9 of 10 (Start time: 08:35:21 PM, 30 December 2024) Total Marks: 1

Which of the following will be used to avoid the problems caused by the BST generated using sorted data?

Select the correct option

☐ Queue

☐ Stack

☒ AVL Tree

☐ Linked list

Reload Math Equations

Click to Save Answer & Move to Next Question

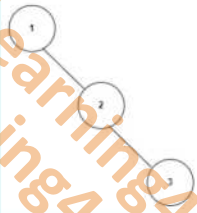
LT [Icons] vulms.vu.edu.pk/Quiz/QuizQuestion.aspx [Icons]

Handouts.pdf Login - Exams MTH603 Mid Term... CS601 Mcqs_Combi... plantuml.com/sitem... Phonetics and Phon... ENG-508 Handouts... Teaching of Listenin...

The extension upgrade requires additional permissions. Go to the extensions manager to accept the upgrade. Go to Extensions

Question # 10 of 10 (Start time: 08:35:59 PM, 30 December 2024) Total Marks: 1

What will be the value of root node if we apply rotation to this non AVL tree?



```
graph TD; 1((1)) --- 2((2)); 1 --- 3((3));
```

Select the correct option

☒ Rotation is not required

☐ 3

☐ 2

☐ 1

Reload Math Equations

Search [Icons] vulms.vu.edu.pk/Quiz/QuizQuestion.aspx [Icons]

Handouts.pdf Login - Exams MTH603 Mid Term... CS601 Mcqs_Combi... plantuml.com/sitem... Phonetics and Phon... ENG-508 Handouts... Teaching of Listenin...

The extension upgrade requires additional permissions. Go to the extensions manager to accept the upgrade. Go to Extensions

CS301P - Data Structures (Practical) (Graded Quiz - 4) Quiz Start Time: 07:43 PM, 07 January 2025

Question # 1 of 10 (Start time: 07:43:28 PM, 07 January 2025) Total Marks: 1

The ASCII value for character 'A' is:

Select the correct option

☒ 65

☐ 66

☐ 68

☐ 67

Reload Math Equations

How many rotations will be made if a node is inserted as left child of the node R in the given tree?

```
graph TD; L((L)) --- P((P)); L --- M((M)); P --- R((R)); M --- T((T));
```

Select the correct option

☐ 4

☒ 2

☐ 1

☐ 3

Reload Math Equations

How many rotations will be made if a node is inserted as left child of the node R in the given tree?

```
graph TD; L((L)) --- P((P)); L --- M((M)); P --- R((R)); M --- T((T));
```

Select the correct option

☐ 4

☒ 2

☐ 1

☐ 3

Reload Math Equations

Handouts.pdf Login - Exams MTH603 Mid Term... CS601 Mcqs_Combi... plantuml.com/sitem... Phonetics and Phon... ENG-508 Handouts... Teaching of Listenin...

The extension upgrade requires additional permissions. Go to the extensions manager to accept the upgrade. Go to Extensions

What will be the balance factor of node 2 if node 1 is deleted from the given tree ?

```
graph TD; 4((4)) --- 2((2)); 4 --- 6((6)); 2 --- 1((1)); 2 --- 3((3)); 6 --- 5((5)); 6 --- 7((7))
```

Select the correct option

Reload Math Equations

☐ 2

☐ 1

☐ 0

☒ -1

Handouts.pdf Login - Exams MTH603 Mid Term... CS601 Mcqs_Combi... plantuml.com/sitem... Phonetics and Phon... ENG-508 Handouts... Teaching of Listenin...

The extension upgrade requires additional permissions. Go to the extensions manager to accept the upgrade. Go to Extensions

What will be the balance factor of node 6 if node 5 is deleted from the given tree ?

```
graph TD; 4((4)) --- 2((2)); 4 --- 6((6)); 2 --- 1((1)); 2 --- 3((3)); 6 --- 5((5)); 6 --- 7((7))
```

Select the correct option

Reload Math Equations

☐ 1

☒ 0

☐ 2

☐ -1

Handouts.pdf Login - Exams MTH603 Mid Term... CS601 Mcqs_Combi... plantuml.com/sitem... Phonetics and Phon... ENG-508 Handouts... Teaching of Listenin...

The extension upgrade requires additional permissions. Go to the extensions manager to accept the upgrade. [Go to Extensions](#)

Question # 5 of 10 (Start time: 07:47:23 PM, 07 January 2025) Total Marks: 1

What is the total frequency of character 'B' in the given Huffman encoding tree?

Select the correct option

☒ 5

☐ 9

☐ 14

☐ 0

[Reload Math Equations](#)

Handouts.pdf Login - Exams MTH603 Mid Term... CS601 Mcqs_Combi... plantuml.com/sitem... Phonetics and Phon... ENG-508 Handouts... Teaching of Listenin...

The extension upgrade requires additional permissions. Go to the extensions manager to accept the upgrade. [Go to Extensions](#)

Question # 6 of 10 (Start time: 07:48:20 PM, 07 January 2025) Total Marks: 1

Which of the following bit stream represents the string 'cab' in the given Huffman tree?

Select the correct option

☐ 110 100 10

☒ 100 0 100

☐ 100 0 10

☐ 100 100 10

[Reload Math Equations](#)

Handouts.pdf Login - Exams MTH603 Mid Term... CS601 Mcqs_Combi... plantuml.com/sitem... Phonetics and Phon... ENG-508 Handouts... Teaching of Listenin...

The extension upgrade requires additional permissions. Go to the extensions manager to accept the upgrade. Go to Extensions

CS301P - Data Structures (Practical) (Graded Quiz - 4) Quiz Start Time: 07:43 PM, 07 January 2025

Question # 7 of 10 (Start time: 07:49:12 PM, 07 January 2025) Total Marks: 1

Which of the following represents frequency of all characters in Huffman Encoding tree?

Select the correct option

Leaf Node

Left Branch

Right Branch

Root Node

Reload Math Equations

Search vulms.vu.edu.pk/Quiz/QuizQuestion.aspx

Handouts.pdf Login - Exams MTH603 Mid Term... CS601 Mcqs_Combi... plantuml.com/sitem... Phonetics and Phon... ENG-508 Handouts... Teaching of Listenin...

The extension upgrade requires additional permissions. Go to the extensions manager to accept the upgrade. Go to Extensions

CS301P - Data Structures (Practical) (Graded Quiz - 4) Quiz Start Time: 07:43 PM, 07 January 2025

Question # 8 of 10 (Start time: 07:49:58 PM, 07 January 2025) Total Marks: 1

If the tree becomes unbalance after deleting a node then we use _____ to rebalance it.

Select the correct option

Heap

Insertions

Stack

Rotations

Reload Math Equations

Search vulms.vu.edu.pk/Quiz/QuizQuestion.aspx

Handouts.pdf Login - Exams MTH603 Mid Term... CS601 Mcqs_Combi... plantuml.com/sitem... Phonetics and Phon... ENG-508 Handouts... Teaching of Listenin...

The extension upgrade requires additional permissions. Go to the extensions manager to accept the upgrade. Go to Extensions

CS301P - Data Structures (Practical) (Graded Quiz - 4) Quiz Start Time: 07:43 PM, 07 January 2025

Question # 8 of 10 (Start time: 07:49:58 PM, 07 January 2025) Total Marks: 1

If the tree becomes unbalance after deleting a node then we use _____ to rebalance it.

Select the correct option

Heap

Insertions

Stack

Rotations

Reload Math Equations

Search vulms.vu.edu.pk/Quiz/QuizQuestion.aspx

vulms.vu.edu.pk/Quiz/QuizQuestion.aspx

Handouts.pdf Login - Exams MTH603 Mid Term... CS601 Mcqs_Combi... plantuml.com/sitem... Phonetics and Phon... ENG-508 Handouts... Teaching of Listenin...

The extension upgrade requires additional permissions. Go to the extensions manager to accept the upgrade. Go to Extensions

CS301P - Data Structures (Practical) (Graded Quiz - 4) Quiz Start Time: 07:43 PM, 07 January 2025

Question # 9 of 10 (Start time: 07:50:24 PM, 07 January 2025) Total Marks: 1

A binary tree will not be considered an AVL tree if the difference between left and right subtree of each node is not more than :

Select the correct option

Reload Math Equations

☐ 0
☐ 3
☒ 1
☐ 2

vulms.vu.edu.pk/Quiz/QuizQuestion.aspx

Handouts.pdf Login - Exams MTH603 Mid Term... CS601 Mcqs_Combi... plantuml.com/sitem... Phonetics and Phon... ENG-508 Handouts... Teaching of Listenin...

The extension upgrade requires additional permissions. Go to the extensions manager to accept the upgrade. Go to Extensions

What is the encoding of character 'C' in the given Huffman encoding tree?

```

graph TD
    14((14)) -- 1 --> 9((9))
    14 -- 0 --> 5((5))
    9 -- 1 --> A((A  
4))
    9 -- 0 --> B((B  
5))
    5 -- 1 --> C((C  
3))
    5 -- 0 --> D((D  
2))
  
```

Select the correct option

Reload Math Equations

☒ 01
☐ 11
☐ 10
☐ 00